# MOHAMMED AATHIL P

**B.E. Electronics and Communication Engineering**
Thiagarajar College of Engineering | MADURAI,TAMILNADU
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## EDUCATION

**Pansy Vidyalaya Matriculation Higher Secondary School** MADURAI
*10th* **2022**
Percentage: **84.4**

**TamilNadu Government Polytechnic college** madurai
*Diploma* **2022-2025**
Percentage: **87.83**

**thiagaraja college of enginreering** madurai
*B.E* **2025-2028**
CGPA: **7.73 (current)**

## AREA OF INTEREST

image processing , robotics , embedded system , remote sensing

## TECHNICAL SKILLS

- **programming languages :** java
- **Automation :** PLC Programming [Ladder Logic]
- **Design / Engineering Tools :** MATLAB ,OrCAD Capture ,Cadence Allegro ,PCB Design, Eagle , AUTO CAD , GNU radio
- **AI / Image Processing :** OpenCV YOLO Image Processing Computer Vision

## PROJECTS

**AI-Based Autonomous Quadruped Robot – Sentinel**
*Academic Project[diploma] / Robotics & Automation Projec*
Designed and developed an autonomous quadruped robot using Arduino Mega and Raspberry Pi 4 for robotic movement, sensor-based navigation and wireless communication. The robot uses 12 servo motors, sensors and HC-12 communication for control and monitoring. Raspberry Pi is used for AI-based functions such as face recognition, while Arduino handles real-time servo control.

**AI-Based Thermo Vision Helmet for Human Detection in Smoke-Dense Environment Project Type:**
*Academic Project[design thinking] / AI & Embedded Systems Project*
Developed an AI-based thermal vision helmet for detecting humans in smoke-dense environments. Integrated a thermal camera, Raspberry Pi, temperature sensors, GPS, RFID and wireless communication. Implemented computer vision and AI-based human detection to assist firefighters during rescue operations.

**Satellite-Based Soil Moisture Monitoring Using Sentinel-1 SAR Data**
*International Conference /Remote Sensing*
Analyzed Sentinel-1 GRD satellite data to estimate soil moisture conditions using VV and VH polarization backscatter values. Performed satellite data preprocessing and analysis using SNAP and generated seasonal soil moisture information for the study area.

**Smart Solar-Powered Drying and Compact Packaging System for Agarbatti Manufacturing**
*hackathon project*
Developed a smart, solar-powered system to support home-based agarbatti manufacturing by rural women artisans. The system provides controlled drying using solar energy and enables compact packaging of finished agarbatti products, reducing manual effort, improving drying efficiency, product quality, and packaging convenience.

## PROFESSIONAL EXPERIENCE

**BSNL** MADURAI
*STUDENT INTERN* **JUNE 2024**
Gained practical exposure to telecommunication systems, networking, optical fiber communication, and basic telecom infrastructure. Learned about the practical operation and maintenance of communication networks.

**INDIAN SPACE ACADEMY** NEW DELHI

*STUDENT INTERN* **JULY 2026**

Worked on satellite-based soil moisture analysis using Sentinel-1 GRD data for the Madurai region. Performed satellite data preprocessing using SNAP, including speckle filtering and terrain correction, followed by analysis of VV and VH polarization backscatter values to estimate and assess soil moisture conditions across different seasons.

## LEADERSHIP & ACTIVITIES

- MATHS CLUB- EVENT CO-ORDINATOR
- SYMPOSIUM - VOLUNTEER AND MEDIA TEAM